

$$f(n, m) = \sum_{i=0}^n \sum_{j=0}^m (n, i) (m, j) g(i, j) \iff g(n, m) = \sum_{i=0}^n \sum_{j=0}^m (-1)^{n+m-i-j} (n, i) (m, j) f(i, j)$$